Airborne Small Target Detection Method Based on Multimodal and Adaptive Feature Fusion

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***Abstract*— The detection of airborne small targets amidst cluttered environments poses significant challenges. Factors such as the susceptibility of a single RGB image to interference from the environment in target detection and the difficulty of retaining small target information in detection necessitate the development of a new method to improve the accuracy and robustness of airborne small target detection. This article proposes a novel approach to achieve this goal by fusing RGB and infrared (IR) images, which is based on the existing fusion strategy with the addition of an attention mechanism. The proposed method employs the YOLO-SA network, which integrates a YOLO model optimized for the downsampling step with an enhanced image set. The fusion strategy employs an early fusion method to retain as much target information as possible for small target detection. To refine the feature extraction process, we introduce the self-adaptive characteristic aggregation fusion (SACAF) module, leveraging spatial and channel attention mechanisms synergistically to focus on crucial feature information. Adaptive weighting ensures effective enhancement of valid features while suppressing irrelevant ones. Experimental results indicate 1.8% and 3.5% improvements in mean average precision (mAP) over the LRAF-Net model and Infusion-Net detection network, respectively. Additionally, ablation studies validate the efficacy of the proposed algorithm's network structure.**

***Index Terms*— Attention mechanism, feature fusion, multimodal, small target detection, unmanned aerial vehicle (UAV) aerial imagery.**

I. Introduction

In recent years, unmanned aerial vehicles (UAVs) have been increasingly utilized in various scenarios and are widely used in military and civil applications [1]. These applications include detection [2], rescue [3], reconnaissance [4], and courier [5], due to their small size, ease of use, and low safety risk factors. Images taken by UAV constitute datasets that can be utilized for research in a multitude of directions [6], [7], [8], [9], [10], [11]. Furthermore, scholars have been engaged in research pertaining to the processing of image aspects [12]. While UAV aerial imagery plays an important role in many fields, accurately identifying and detecting small targets remain a challenge. Target detection has a wide range of applications on UAV platforms, including emergency rescue, border patrol, mapping and assessment, and environmental monitoring. However, due to the elevated flight altitude of UAVs, the diminutive scale of small targets in aerial images, the dearth of detailed information regarding their appearance, and the limited number of useful features that can be extracted, there is a heightened necessity for the precise localization of small targets. The issue of a vast number of minute targets and the deficiency in the capacity to accurately define their characteristics in aerial images have had a considerable impact on the efficacy of small target detection. UAVs typically use on-board cameras to capture aerial imagery for target detection; however, RGB images contrast and detail are lost in low light conditions, making it difficult to extract effective features, which is unreliable for missions involving personnel safety. In addition, the detection of small targets in aerial imagery is often affected by lighting conditions, target size, background clutter, and other factors, creating many limitations and challenges for traditional target detection methods. The integration of UAVs with target detection technologies has allowed their unique advantages to emerge. However, target detection by UAVs still faces a series of challenges in various complex real-world scenarios.

Convolutional neural networks (CNNs) have demonstrated significant advancement in image classification [13], [14], image denoising [15], hyperspectral unmixing [16], [17], super-resolution restoration [18], [19], and change detection [20], [21], [22], [23], [24]. In image fusion, CNNs have also been used to implement multifocal image fusion, multiscale transforms, and deep Boltzmann machines with remarkable results. Liu et al. [25] proposed a CNN-based multifocal image fusion scheme. They trained a Siamese network [26] to classify focused and unfocused image blocks and obtained accurate binary decision maps. The fusion image is created by combining the decision map with the corresponding

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source image. Tang et al. [27] improved the algorithm and proposed a pixel-level CNN. Wu et al. [28] proposed a framework for fusing infrared (IR) and RGB images using multiscale transforms and deep Boltzmann machines. The R-CNN family of algorithms, as described in [29], generates a substantial number of candidate frames through a selective search model. These candidate frames are then subjected to target classification and position regression, with the objective of achieving high target recognition accuracy. However, due to the overlap between candidate frames, a considerable amount of redundant information is generated, which has the effect of reducing the efficiency of target detection. To address this issue, subsequent methods such as Fast R-CNN [30] and Faster R-CNN [31] have been proposed. The Fast R-CNN algorithm performs a single convolutional operation on the entire image to extract features through the shared convolution technique, which improves efficiency, and the Faster R-CNN algorithm significantly reduces the computational time of the model. The regional proposal network (RPN) is employed in lieu of the selective search model, which further reduces the generation of redundant candidate frames and further improves the efficiency of the algorithm. Nevertheless, the aforementioned two methods are susceptible to the inadvertent omission of small target elements within the image, thereby compromising the integrity of the captured data.

CNN-based methods for target detection are increasingly popular. Among these methods, the YOLO family of algorithms is known for its fast detection capability. The initial version of the YOLO algorithm [32] predicts the category and location of all targets in a single inference by segmenting the image into multiple grids, which achieves fast detection but low accuracy in small target detection. In 2017, to improve the recognition of small targets, the YOLOv2 algorithm [33] was developed. This algorithm improves the recognition of small targets by introducing anchor boxes and using K-means cluster analysis to determine the most suitable anchor size for the dataset. Furthermore, this iteration enhances the detection of small targets by fusing different levels of features. Subsequently, YOLOv3 [34] incorporated multiscale detection into v2, utilizing a more robust Darknet-53 network as its foundation to enhance accuracy and sustain real-time performance. YOLOv4 [35] was introduced in 2020, integrating a multitude of cutting-edge techniques and optimization strategies to achieve a more balanced relationship between speed and accuracy. In the same year, 2020, Ultralytics introduced YOLOv5 [36], which was further optimized on the basis of v4 by introducing the spatial pyramid pooling fast (SPPF) structure to replace the traditional spatial pyramid pooling (SPP), making the application scenarios even more extensive. In 2021, the YOLOX algorithm [37] further simplified the model structure by introducing the anchor-free mode, while the YOLOR algorithm [38] presented the multitasking network for simultaneous target detection and segmentation. In 2022, YOLOv6 [39] demonstrated an improvement in performance through the introduction of the RepVGG structure and the integration of multibranching with residual connection optimization. In contrast, YOLOv7 [40] optimized the model architecture and dynamic label assignment strategy.

Liu et al. [41] proposed the single shot multibox detector (SSD) algorithm, which combines the advantages of Faster R-CNN and YOLO. The SSD model draws upon the core concepts of Faster R-CNN, establishes multiple sizes of a priori frames, and employs multiscale feature maps for prediction. The method achieves a satisfactory comprehensive detection effect by fusing the prediction frames of different scale feature maps. Fu et al. [42] introduced the deconvolutional single shot detector (DSSD) algorithm, which employs ResNet as the core backbone network and adds a residual module before performing classification and regression operations. The method effectively combines deep and shallow features, thereby improving the detection accuracy of the model through the construction of an hourglass-type structure. Furthermore, the feature fusion SSD (FSSD) [43] algorithm employs a parallel approach to combine feature layers of varying scales, subsequently generating new feature pyramid layers through downsampling techniques with the objective of enhancing the detection performance.

Despite the considerable success of the YOLO series models in a number of fields, including target detection, where they have been used with great effect, there are still a number of challenges to be overcome. However, in the context of target detection from an UAV perspective, the challenge currently facing researchers is to achieve an optimal balance between detection accuracy and model scale. In order to address this issue, a number of researchers have proposed a series of innovative approaches to enhance the YOLO model. For instance, Lou et al. [44] have developed a new downsampling method and modified the feature fusion network of YOLOv8 in order to enhance the detection capability of dense small targets. Similarly, Guo et al. [45] have designed a dense connectivity mechanism and proposed a new network structure module, C3D, in order to retain the information more comprehensively. Wang et al. [46] incorporated a small target detection structure, STC, into the neck of YOLOv8 and introduced a global attention mechanism (GAM), to minimize the loss of feature information. Furthermore, Wang et al. [47] integrated the BiFormer attention mechanism into the backbone network, resulting in the focal FasterNet block (FFNB) feature processing module. This module introduced two new detection scales, significantly reducing the leakage rate. Nevertheless, for certain small objects, such as bicycles, the existing detection accuracy of these methods still requires improvement.

The algorithms for target detection described above primarily utilize RGB images for training and detection purposes. However, in real-world environments, RGB imaging is susceptible to changes in lighting and weather conditions. Therefore, in unfavorable conditions such as low visibility, low illumination, and nighttime, target detection methods based solely on RGB images are ineffective.

IR sensors, which utilize thermal radiation for imaging, are not affected by lighting conditions, thereby offering significant advantages for target detection. Consequently, it is possible to fuse RGB images with IR image information. IR images have the advantage of reducing interference from external factors such as sunlight and fog, and they can easily distinguish

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the target from the background. IR images are frequently used to detect and recognize low-resolution objects [48], [49]. Meanwhile, RGB images contain numerous visible edges and object details that match human visual characteristics, resulting in better object detection performance [50]. Several studies have extensively researched RGB-IR multimodal target detection. In their study, Li et al. [51] proposed a deep learning-based feature fusion technique that is designed to differentiate between the basic and detailed levels of the input image. The method employs a weighted average technique with an L1-paradigm fusion strategy to generate a multilevel fused image, from which the maximum result is selected as the final fused image. Furthermore, in 2019, the DenseFuse network proposed by Li and Wu [52] extracted and fused RGB and IR image features through the codec structure, successfully reconstructing a high-quality fused image, which demonstrated excellent results. In order to investigate the impact of different stages of network layer fusion on detection effectiveness, Hwang et al. [53] constructed the Korea Advanced Institute of Science and Technology (KAIST) dataset, which encompasses a diverse range of RGB and IR image environments. They then developed an innovative multimodal feature fusion method (ACF + T + THOG, the ACF is the characteristic defined for the color channel, and T + THOG indicates additional channel characteristics that have been enhanced from the hot channel), which is based on traditional feature extraction. However, despite this, the target recognition effect did not meet expectations. Feichtenhofer et al. [54] proposed a novel concept that the fusion of images at different stages of the deep learning framework can more effectively extract feature information from the original image, thereby enabling the learning of complex data structures. In the same year, Wagner et al. [55] and Liu et al. [56] proposed a multimodal pedestrian target detection method based on CNN fusion. In this study, Wagner et al. [55] compared two CNN fusion strategies, early and late, and found that late fusion was more effective. Liu et al. [56] tested several fusion approaches combining two-branch CNNs at different stages, and the experimental results showed that mid-stage fusion performed the best. Li et al. [57] used the multispectral simultaneous detection and segmentation (MSDS) in pedestrian detection and semantic segmentation tasks using the MSDS-RCNN's dual-stream mid-stream fusion framework, which significantly improved the accuracy of detection.

It can be reasonably concluded, therefore, that the fusion of RGB and IR images represents an optimal approach for UAV detection missions in complex environments. Nevertheless, the current multimodal fusion approach encounters significant challenges in the domain of target detection. One such challenge is to fully consider the impact of different stages on the task during the fusion process and to effectively address the potential for information loss. While numerous existing methods are capable of fusing RGB and IR images, they often fail to adequately address the potential alterations in features across different fusion stages, leading to suboptimal fusion outcomes. In particular, the fusion process is prone to information loss when detecting small targets, due to their relatively small size and sparse features. This further underscores the limitations of current fusion methods. Furthermore, due to the disparity in imaging principles, there are notable regional discrepancies between RGB images and IR images. RGB images predominantly reflect light with high texture and gradient variations, whereas IR images convey thermal radiation information with high contrast. In conditions of low light or at night, RGB images are unable to obtain information about target features. Conversely, IR images offer a valuable addition to RGB images, as they possess superior interference immunity and are able to clearly recognize targets even in low-light conditions. Further study is required to determine the optimal method for extracting fused features from the two modalities while retaining richer and more effective information.

This article aims to enhance the robustness and accuracy of UAVs for small target detection tasks under complex lighting and weather conditions. A self-adaptive characteristic aggregation fusion (SACAF) module is proposed to accurately focus on important feature information and extract and enhance beneficial features between modalities. The article presents the following main contributions.

1) This article proposes the YOLO-SA network for the detection of airborne small targets. The network comprises a feature fusion module and a YOLO network. Given the distinctive characteristics of RGB and IR images, the fusion of the two was deemed an optimal approach for small target detection. The network addresses the issue of small target features being easily lost within the network by employing an early fusion strategy to retain as much target information as possible prior to the feature map entering the extraction network. Additionally, it incorporates dual attention to accommodate small target detection.
2) In order to fully exploit the benefits of the complementary bimodal features in the early fusion process, we propose a SACAF module that employs the dual-attention mechanism to enhance the features of each modality and highlight the key features. Furthermore, an adaptive weighting formula is developed to effectively enhance the validity of features and suppress the invalidity of features.
3) In order to verify the performance of our proposed network structure, we have undertaken a series of data processing operations. We cleaned the DroneVehicle dataset [58] and corrected the mislabeled parts of it, and reclassified the KAIST dataset and LLVIP dataset [59] into training and test sets. All three processed datasets existed to meet the definition of a small target, and finally the effectiveness of the proposed method was verified experimentally.

This article is structured as follows. Section II discusses the theoretical related work on small target detection. Section III describes our proposed method in detail. Section IV contains experimental details and results. Finally, Section V gives broad conclusions and future research direction.

II. Preliminaries

In contrast to the classical sliding window strategy and RPNs, the YOLO algorithm employs an image-wide approach

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to target detection. It treats the detection process as a regression and classification problem, predicting the location (bounding box) and the class to which the target belongs directly from the features of the image. In the YOLO workflow, the image features are initially extracted by a CNN, then fused, and finally categorized and localized at the prediction layer. This process enables the determination of the target's precise position and confidence in the predicted category. The algorithm first divides the image to be detected into a grid of s × s cells, each of which can detect the object whose center point falls within the grid cell. Each cell will have multiple bounding boxes and a confidence score for the target category.

The YOLO network utilized in this article is based on version 6.0 of YOLOv5, which has undergone fine-tuning from version 5.0 to incorporate a multitude of novel features. The high adaptability of this version has enabled the model to be applied to a wider range of applications than other YOLO networks. Furthermore, the model's performance and stability have been comprehensively optimized and improved. The emergence of YOLOv5 version 6.0 has brought a new milestone in the field of real-time target detection. The network architecture of this version encompasses the input layer, backbone network, header network, and detection layer. In the version, the ConvBNSiLU (CBS) structure replaces the original ConvBNLeakyReLU (CBL) module with SiLU activation functions, which are distinguished by the utilization of LeakyReLU activation functions. Concurrently, a structure configured as CBS (kernel_size = 6, stride = 2, padding = 2) is introduced in the initial layer, replacing the Focus layer, with the objective of optimizing the model initialization process. To streamline the model architecture, the SPPF layer was selected to replace the SPP layer, thereby reducing the number of computational steps. To enhance processing speed, the number of C3 modules in the backbone network was reduced from nine to six, and a shortcut connection was added to the final C3 module of the network, which is connected to the SPPF layer. Furthermore, the hyperparameters were adjusted and novel data enhancement strategies, such as mixup and copy-paste, were introduced. In addition to the standard four network dimensions (small, medium, large, and extra-large), two smaller model variants, YOLOv5n and YOLOv5n6, have been incorporated into version 6.0 of the model. These variants are consistent with the main model architecture.

This article aims to detect small targets. However, the criteria for defining small targets vary from scenario to scenario. Academics typically categorize the definition into two main categories: based on relative scale and based on absolute scale. In academic research, the definition of relative scale-based small targets usually involves the relative ratio of the target to the image. Small targets are defined based on their absolute pixel size, with the most common standard being those with a resolution of less than 32 pixels × 32 pixels, as defined by the MS COCO dataset in the field of target detection. This article uses targets smaller than 32 pixels × 32 pixels as the criterion for defining small targets.

In the exploration of previous work [55], [60], [61], it can be summarized that the fusion of RGB image and IR image information can be divided into three modes: early fusion, mid-term fusion, and late fusion [62]. Early fusion allows for the full exploitation of multimodal information, although this may result in information redundancy. Mid-term fusion enables the complete integration of information at the feature extraction stage, but this comes at the cost of higher computational complexity. Late fusion preserves more modal information, but there is a risk of information loss. Accordingly, in order to enhance the accuracy of small target detection, this article proposes the strategy of early fusion, which is combined with the purpose of improving the accuracy of small target detection. Furthermore, in order to mitigate the impact of noise and background on small target detection, the adaptive feature fusion module is introduced to deeply mine and effectively utilize the key features in RGB and IR images. This is done in order to enhance the expressive power of the features at the global level.

III. Proposed Method

Considering recent research advances in feature fusion mechanisms, we propose a novel network for detecting airborne small targets that utilize multimodal feature fusion to improve accuracy. The network architecture for small-target detection is discussed in detail in Section III-A, the adaptive feature fusion module is explained in Section III-B, and the loss function for network training is described in Section III-B1.

A. Model Architecture

Fig. 1 depicts the structural model of the YOLO-SA network. The primary structure comprises an image input and a YOLO network structure with a feature fusion module. This method initially performs preliminary feature extraction on two distinct modal images. Following this, the obtained feature maps are fed into the SACAF module within the YOLO network for analysis. This module is capable of resolving the issues associated with small targets, which are prone to loss of information and exhibit low detection accuracy during the target detection process. The module initially employs the spatial attention mechanism to compute the respective weight matrices of the two feature maps, after which a specific weight formula is utilized to synthesize the enhanced image features. The enhanced feature map is then processed by channel attention to clarify the correlation between different channels. At the same time, the weights of each modal feature information are adjusted to highlight the target features and suppress the background noise, thus enabling the enhancement of features. Subsequently, the fused and enhanced feature maps are fed into the backbone network of the adapted YOLO. These feature maps are then passed to the head, which is downsampled three times to perform the task of target classification and localization.

B. Self-Adaptive Characteristic Aggregation Fusion

The algorithm employs a multimodal fusion approach to enhance the performance of small target detection in UAV aerial images. To this end, an innovative adaptive feature fusion module is proposed, based on the earlier fusion strategy.

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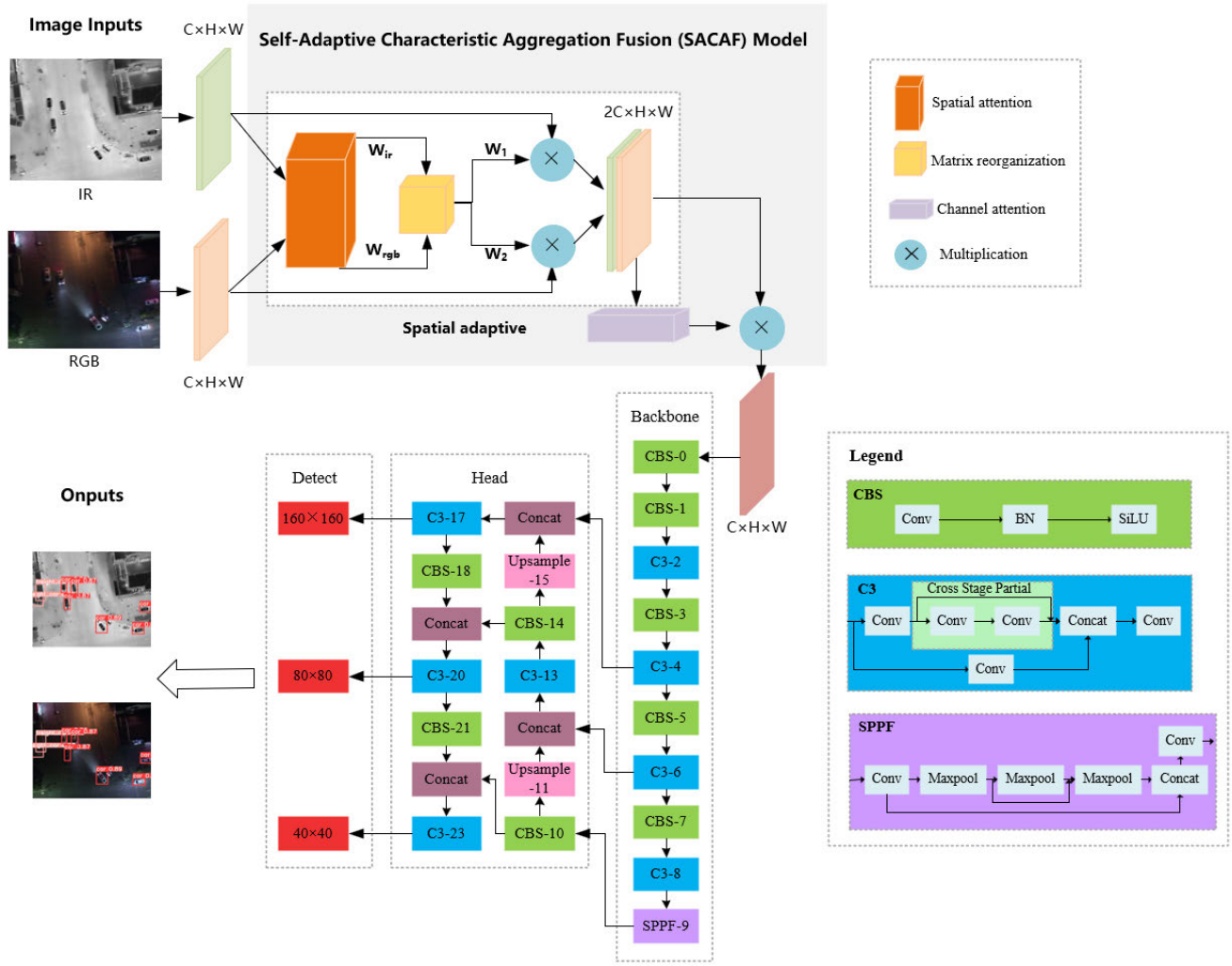


Fig. 1. Illustration of the proposed YOLO-SA network, where the gray block is the SACAF module proposed in this article and the lower half of the model is the YOLO network used. The spatial adaptative component is delineated in Fig. 2.

As illustrated in Fig. 1, the SACAF module effectively fuses the features of the initially extracted RGB and IR images. It enhances the target features while retaining the small target information as much as possible, and utilizes the dual-attention mechanism in the fusion process to enhance the features of each modality and highlight the key features. In addition, an adaptive weighting formula is proposed to effectively enhance the effective features and suppress the invalid features.

1) Dual-Attention Mechanism: The use of the attention mechanism is crucial in retaining effective information from the feature map, particularly for targets that represent a small proportion of the feature map. This mechanism highlights effective features and reduces information loss during feature extraction. Spatial attention, channel attention, and dual-attention mechanism each have unique advantages and are widely used in various target detection tasks, particularly in areas involving small target detection.

The dual-attention mechanism is a lightweight universal module that enhances the network's representational power without significantly increasing its parameters. The dual-attention mechanism assigns weights to the individual channels of the feature map and considers the weight parameters and feature information of the spatial part. Bottleneck attention module (BAM) and convolutional block attention module (CBAM) are hybrid domain attention mechanisms that combine channel attention and spatial attention. BAM does this through parallelism, while CBAM does it sequentially. This subtle difference reflects the distinct strategies and optimization paths of the two mechanisms in processing complex visual information. As a result, they provide more flexible and effective solutions in practical applications.

The SACAF module is proposed in this article, building on the inspiration of the dual-attention mechanism mentioned earlier and combined with the detection task. The SACAF module optimizes the feature extraction process by utilizing both spatial and channel attention mechanisms. The precise methodology is as follows.

The feature maps of the two modalities are fed into the spatial attention module separately. The specific steps of the spatial attention module are depicted in Fig. 2. In the spatial attention module, global mean pooling and global maximum pooling operations are first performed on each feature map to obtain the average and maximum values of each spatial

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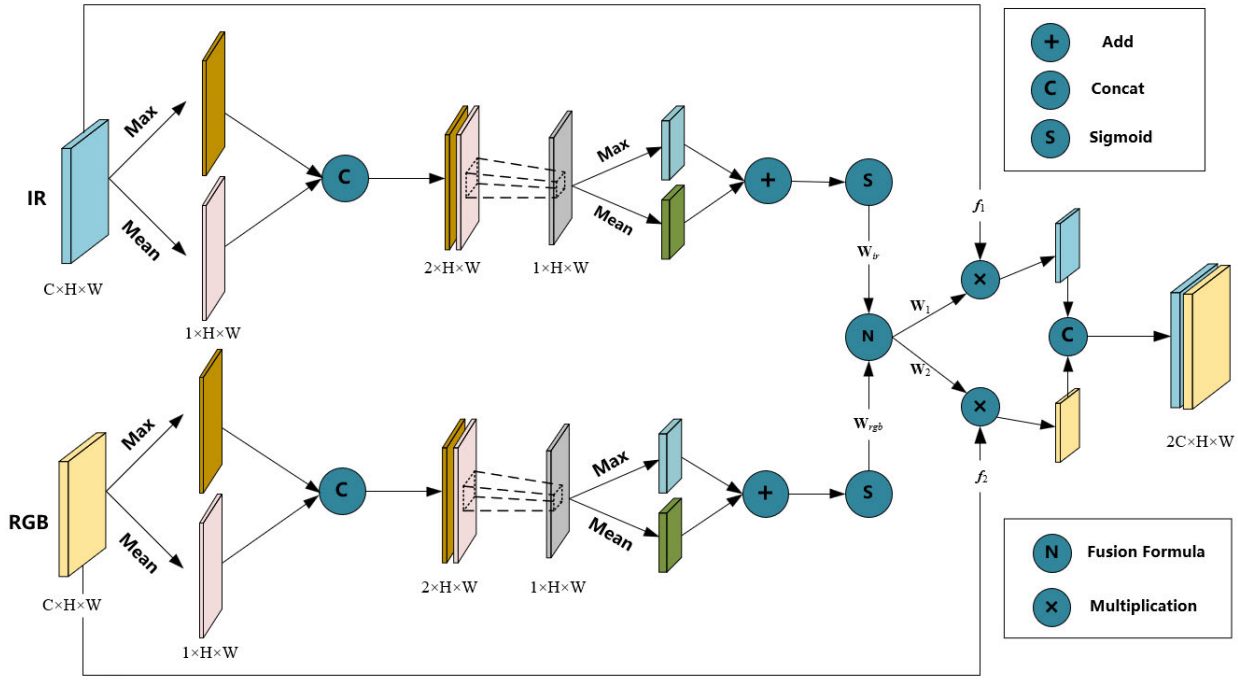


Fig. 2. Spatially adaptive internal structure diagram.

location. Two matrices are then obtained. Second, these two matrices are spliced, and a convolutional layer and sigmoid function are used to learn the weights of each spatial location. The resulting weight matrix is then multiplied with the original feature map and fused by a weight formula to produce features with enhanced spatial importance. Finally, this new fused feature map is fed into the channel attention module to further optimize the feature extraction and fusion process.

In the channel attention module, the input feature maps are first subjected to global average pooling and global maximum pooling operations, and then the results of the two poolings are spliced according to the channel. Next, the spliced feature maps are fed into the fully connected layer for predicting the importance of each channel and obtaining the weights of each channel. Finally, these channel weights are applied to the initial feature map through matrix operations to generate the enhanced feature map.

In this step, the spatial attention mechanism plays a pivotal role in effectively suppressing noise and other irrelevant information in the feature map, thus highlighting important features. In contrast, the channel attention module focuses on retaining the salient target feature information in the channel, further enhancing the accuracy and efficiency of target detection. The design of this serial structure not only improves the quality of feature extraction but also ensures the efficiency of the processing.

The application of the SACAF module demonstrates the synergistic effect of spatial and channel attention mechanisms in target detection tasks. This innovative approach has the potential to improve model performance, particularly when dealing with complex multimodal data, as the SACAF module is able to identify and process key information more accurately, resulting in more precise and reliable target detection in a variety of application scenarios.

2) Adaptive Weighting: While the ordinary attention matrix can be effective for target enhancement, it is important to fully utilize the effective features of different modalities in multimodal tasks to improve target detection performance. For instance, RGB images' color and texture features are more crucial in well-lit scenes, whereas IR images' contour information is more important in dark scenes or poorly illuminated scenes with occlusions. Therefore, a new fusion method is required to balance the fusion process.

In the feedforward process of CNNs, a single convolutional kernel uses the same set of weights for the entire input data. However, it is particularly important to consider the variability of feature information of different inputs in a two-input network model. The SACAF module combines the spatial and channel attention mechanisms to adaptively assign weights by customizing them.

The SACAF module takes two feature maps, one for IR and one for RGB images, both with dimensions $C \times H \times W$, as input. The spatial attention module processes these feature maps to obtain the corresponding weight matrices, $\mathbf{W}_{\text{ir}}$ and $\mathbf{W}_{\text{rgb}}$. These weight matrices are then used in the following equation to generate a new weight matrix:

$$\begin{cases} \mathbf{W}_1 = \text{soft}\max\left(\mathbf{W}_{\text{ir}} / \text{norm}\left(\mathbf{W}_{\text{ir}}, \mathbf{W}_{\text{rgb}}\right)\right) \\ \mathbf{W}_2 = \text{soft}\max\left(\mathbf{W}_{\text{rgb}} / \text{norm}\left(\mathbf{W}_{\text{ir}}, \mathbf{W}_{\text{rgb}}\right)\right) \end{cases} \tag{1}$$

where $\mathbf{W}_{\text{ir}}$ and $\mathbf{W}_{\text{rgb}}$ denote the weight matrices of IR features and RGB features obtained after spatial convolution, respectively. $\mathbf{W}_1$ and $\mathbf{W}_2$ denote the new weight matrices obtained after the transformation of these two weight matrices,

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respectively

$$\mathbf{X} = \boldsymbol{f}_1 * \mathbf{W}_1 + \boldsymbol{f}_2 * \mathbf{W}_2 \tag{2}$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ denote the new weight matrices obtained after the transformation of the above two formulas. $\boldsymbol{f}_1$ and $\boldsymbol{f}_2$ denote the IR feature and RGB feature maps, respectively, and $\mathbf{X}$ denotes the new feature matrix finally obtained after the operation of (2).

The method employs a weight matrix adjustment ratio to enhance target information. This ratio is applied to a normalized matrix of two modalities, effectively suppressing background and noise while facilitating more effective feature fusion.

C. YOLO Network

The backbone network structure of the YOLO system can be divided into three layers within the YOLO-SA network. backbone, head, and detect. The image size after several downsampling in the original YOLOv5 model is 20 × 20, and the detector heads output after feature pyramid network (FPN) upsampling are the output detectors, which are 80 × 80, 40 × 40, and 20 × 20, respectively. Of these, 80 × 80 is used to detect small targets, while 20 × 20 is used to detect large targets. In the event that the targets in question are smaller than 8 × 8 pixels, it becomes challenging to discern their feature information through the detection head. Consequently, in order to enhance the detection performance of small targets of varying sizes, our model modifies the downsampling step, resulting in the output detection heads being 160 × 160, 80 × 80, and 40 × 40. Accordingly, the model's multiscale learning capability is enhanced, enabling it to more effectively cope with the presence of a large number of small targets, thereby improving its detection capability in dense scenes.

D. Loss Function

The algorithm employs the loss function of YOLOv5 due to its superior convergence speed and enhanced performance. The loss function is defined as follows:

$$\begin{aligned} L_{\text{total}} &= \sum_{i}^{N} (\lambda_1 L_{\text{box}} + \lambda_2 L_{\text{obj}} + \lambda_3 L_{\text{cls}}) \\ &= \sum_{i}^{N} \left(\lambda_1 \sum_{j}^{B_i} L_{\text{CIoU}_j} + \lambda_2 \sum_{j}^{S_i \times S_i} L_{\text{obj}_j} + \lambda_3 \sum_{j}^{B_i} L_{\text{cls}_j} \right) \end{aligned} \tag{3}$$

where N represents the number of detection layers, while B refers to the number of targets assigned to each a priori box, and i and j represent the grid indexes in the feature map with values in the range of $[0, N]$; furthermore, $S \times S$ denotes the number of grids segmented by the network, the loss functions used in this study include localization loss (L_{box}), confidence loss (L_{obj}), and classification loss (L_{cls}). The weight parameters for L_{box}, L_{obj}, and L_{cls} are represented by λ_1, λ_2, and λ_3, respectively.

YOLOv5 uses complete intersection over union (CIoU) loss [63] as the L_{box}. Then the CIoU loss function can be defined as



Fig. 3. Distance between the target frame and the center of the detection frame.

$$\begin{cases} L_{\text{CIoU}} = 1 - \text{IoU} + \dfrac{\rho^2(\mathbf{b}, \mathbf{b}^{\text{gt}})}{c^2} + \alpha v \\ \text{IoU} = \dfrac{|B \cap B^{\text{gt}}|}{|B \cup B^{\text{gt}}|} \\ v = \dfrac{4}{\pi^2} \left(\arctan \dfrac{w^{\text{gt}}}{h^{\text{gt}}} - \arctan \dfrac{w}{h} \right)^2 \end{cases} \tag{4}$$

where $B^{\text{gt}} = (x^{\text{gt}}, y^{\text{gt}}, w^{\text{gt}}, h^{\text{gt}})$ is the ground truth, and $B = (x, y, w, h)$ is the predicted box. In this context, $(x^{\text{gt}}, y^{\text{gt}})$ and (x, y) represents the coordinates of the upper left corner of the true marker box and the predicted box, respectively. $\mathbf{b}$ and $\mathbf{b}^{\text{gt}}$ denote the central points of B and B^{gt}, respectively. Here, w^{gt} and h^{gt} refer to the width and height of the real labeling box, while w and h are the width and height of the prediction box, respectively. The symbol ρ is used to denote the distance between the centers of the two frames calculated, i.e., the value of d in Fig. 3. And c denotes the farthest distance between the edges of the two boxes. In addition, α is a weighting factor used to adjust the relative importance of different components in the overall assessment.

Both L_{obj} and L_{cls} use BCEWithLogitsLoss, which is calculated as follows:

$$\text{Loss} = \frac{1}{n} \sum_{i}^{n} \left[y_i \cdot \log(\sigma(x_i)) + (1 - y_i) \cdot \log(1 - \sigma(x_i)) \right] \tag{5}$$

where $\sigma(a) = (1/(1 + \exp(-a)))$ is the Sigmoid function. The loss function uses n to denote the size of the binary classification problem, where y_i is the binary label of each sample $y_i \in (0, 1)$, and x_i denotes the probability of that sample $x_i \in [0, 1]$.

In the target detection task, L_{obj} evaluates the network's ability to recognize the presence of a target in the grid, while L_{cls} evaluates the network's accuracy in classifying the target. The network can be effectively trained to optimize for both tasks by using BCEWithLogitsLoss, thus improving overall detection performance.

IV. Experiments

This section presents a quantitative and qualitative evaluation of the proposed method for detecting small targets in

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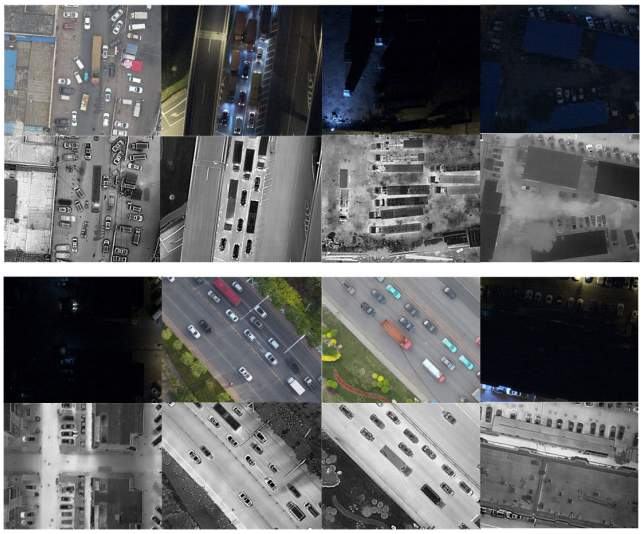


Fig. 4. DroneVehicle dataset partial image pairs.



Fig. 5. KAIST dataset partial image pairs.

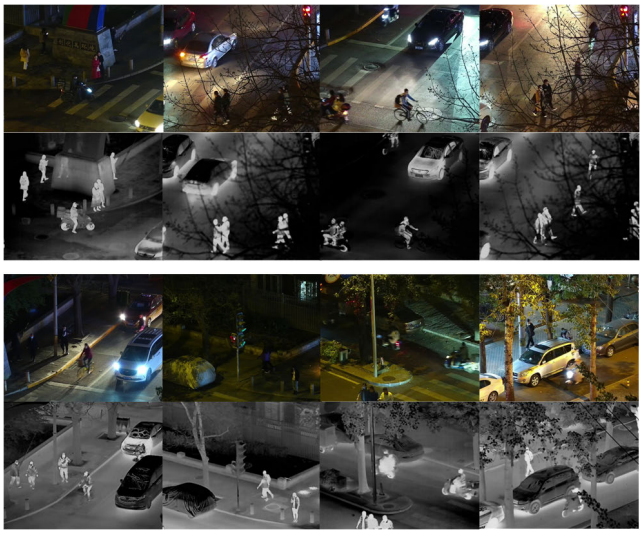


Fig. 6. LLVIP dataset partial image pairs.

airborne scenarios. The evaluation is based on multimodal and adaptive feature fusion and is conducted on three datasets: DroneVehicle, LLVIP, and KAIST.

A. Dataset Description

The DroneVehicle dataset is a comprehensive collection of aerial images captured by UAVs and labeled by Tianjin University. The drone remote sensing dataset comprises 28 439 pairs of RGB-IR images, which have been categorized as vehicles and captured by camera-equipped drones. The regional scenes have been classified as urban roads, residential areas, and highways. The lighting conditions during shooting have been categorized as night and day. The shooting heights were classified as 80, 100, and 120 m, while the shooting angles were categorized as 15°, 35°, and 45°. During the calibration phase of the dataset, the RGB-IR image pairs were cropped using affine transformation and region cropping techniques to ensure that the majority of the cross-modal image pairs were aligned. Fig. 4 illustrates a selection of images from the DroneVehicle dataset. The first row of each set of graphs shows the RGB images and the second row shows the IR images.

The KAIST dataset was proposed by Hwang et al. [53]. It was acquired by an in-vehicle camera and the images of both modalities were subjected to a rigorous alignment process. The dataset comprises images of pedestrian targets captured in a variety of urban settings, including campus and city roads. The image size is 640 × 512. The training set comprises 8963 pairs of RGB and IR images, while the test set contains 2252 pairs of RGB and IR images. Fig. 5 presents a selection of images from the KAIST dataset. The first row of each set of graphs shows the RGB images and the second row shows the IR images.

The LLVIP dataset, proposed by the experimental team of Beijing University of Posts and Telecommunications, employs a binocular camera and comprises 16 836 image pairs, each of which is rigorously aligned. As the majority of the images in this dataset were captured in a dark night environment, they are particularly suited to the assessment of the algorithm's performance in such conditions. Fig. 6 illustrates a selection of image pairs from the LLVIP dataset. The first row of each set of graphs shows the RGB images and the second row shows the IR images.

B. Experimental Setup

1) Parameter Settings: The experiments were conducted using the experimental configuration outlined in Table I. In consideration of the memory limitations of the experimental equipment and the duration of the experiment, the epoch is set to 16, the batch size is chosen to be 4, and the initial learning rate is set to 0.0001.

2) Evaluation Metrics: In order to comprehensively evaluate the performance of the algorithm for multimodal image fusion in the target detection task, this study employs two analysis methods, including subjective and objective. A preliminary performance judgment is obtained through a manual

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TABLE I
EXPERIMENTAL CONFIGURATION

Environment Configuration	Information
CPU	Intel i5-8400
GPU	GeForce RTX 3090(24G Video Memory)
Operating System	Windows10
Development Language	Python 3.8.3
Cuda	Cuda 11.1

and intuitive evaluation of the detection results of selected image samples for the subjective analysis. For objective analysis, a variety of evaluation metrics are employed to provide a comprehensive and detailed quantitative analysis of the algorithm's performance on the test set. This approach aims to provide an accurate and reliable evaluation. In particular, when dealing with the binary classification problem, the prediction results of the image targets are explicitly categorized into both positive and negative classes. This categorization can be demonstrated in detail by constructing a confusion matrix. This evaluation method, which integrates subjective intuition with objective quantification, facilitates a comprehensive understanding of the efficacy of multimodal fusion techniques in the domain of target detection.

In the context of binary classification, the combination of the predicted and real situation dimensions yields four classification results.

1) *True positive (TP):* The number of samples that are correctly identified as belonging to the positive class.
2) *True negative (TN):* The number of samples that are truly negative and correctly identified as such by the model.
3) *False positive (FP):* The number of samples that are, in fact, negative but have been misidentified as positive by the model.
4) *False negative (FN):* The number of samples that are, in fact, positive but are misclassified as negative by the model.

The aforementioned categorization of cases permits the calculation of the principal evaluation metrics utilized in the experiment: the log-average miss rate (MR^{-2}), the average precision (AP), and the mean AP (mAP).

The average miss rate metric employed in this experiment is MR^{-2}, which is utilized as a gauge of multimodal pedestrian detection performance and was proposed by Dollar et al. [69] This metric is derived from the miss rate-false positive per image (MR-FPPI) curve, where MR represents the missing rate as the vertical coordinate of the MR-FPPI curve. The missing rate is calculated as follows:

$$\mathrm{MR} = \frac{\mathrm{FN}}{\mathrm{TP} + \mathrm{FN}} \tag{6}$$

where TP + FN reflects the total number of positive samples in the dataset, and FN denotes the number of those samples that were misclassified by the model as negative samples that were, in fact, positive

$$\mathrm{FPPI} = \frac{\mathrm{FP}}{N_{\mathrm{images}}} \tag{7}$$

where FP is the number of false positive examples in all images and N_{images} is the total number of images.

The process of calculating MR-2 involves selecting nine equally spaced FPPI points in the interval [0.01, 1.0] on a logarithmic scale and averaging the MR corresponding to these points. Consequently, a reduction in the MR^{-2} value is indicative of a decline in the number of missed detections, thereby reflecting the algorithm's enhanced performance in target detection and recognition. This evaluation method not only provides an accurate quantification of the model's performance, but also facilitates the comparison of the relative effectiveness between different algorithms.

The calculation of AP is analogous to the calculation of MR^{-2} values and is also based on the precision–recall (P–R) curve. AP is the average of the corresponding precision P at each level of recall R, where precision P is the proportion of samples that truly belong to the positive class out of all samples predicted to be in the positive class. The formula is given in the following equation:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{8}$$

where R denotes the recall, the number of samples that are correctly predicted out of the actual positive samples, calculated in the following equation:

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{9}$$

The formula for calculating AP is as follows:

$$\mathrm{AP} = \int_0^1 P(R)\,dR \tag{10}$$

where the higher the AP value, the better the performance of the algorithm target detection and recognition.

The mAP is a metric frequently employed in the evaluation of object detection models. It is the arithmetic mean of the AP values for each category. The calculation formula is provided in the following equation:

$$\mathrm{mAP} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{AP}_i \tag{11}$$

where N is the number of categories and $\mathrm{AP_i}$ is the AP of the ith category.

C. Ablation Study

A series of tests were conducted on the DroneVehicle dataset, with the objective of comparing the performance of the unimodal network, the bimodal fusion network, and the bimodal adaptive fusion network in terms of leakage detection rate metrics. Additionally, the performance of the network with the addition of the SACAF module on both the missed detection rate and average accuracy metrics was evaluated. The experimental results were then objectively analyzed to determine the fusion structure with the optimal recognition performance.

The experiments analyzed different fusion methods in depth using the YOLOv5 network model. These fusion methods include unimodal RGB with Y, IR with Y, and bimodal

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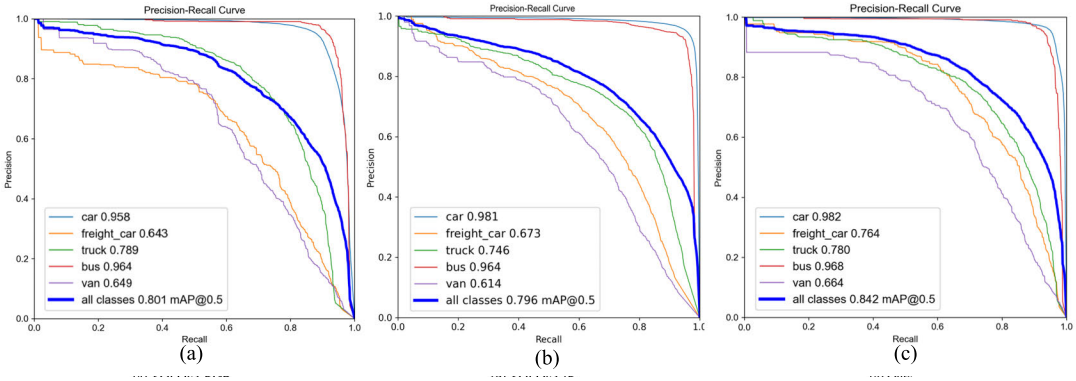


Fig. 7. P–R curves under different models. (a) YOLOv5-RGB. (b) YOLOv5-IR. (c) Ours.

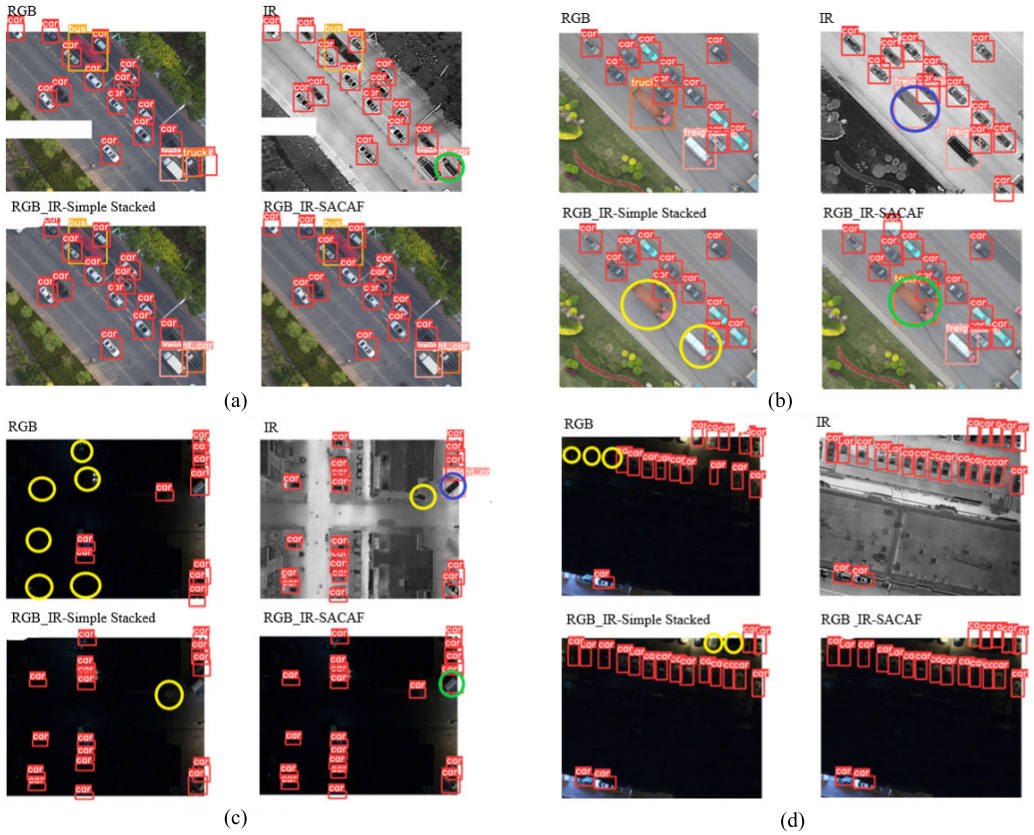


Fig. 8. Plot of different network detection and identification results. The yellow circle indicates a missed target, while the blue and green circles indicate a misdetected target. The blue circle represents a misdetected wrong target, while the green circle represents a misdetected correct target. The detection category is indicated on the box by a marking. (a) Good lighting 1. (b) Good lighting 2. (c) Poor lighting 1. (d) Poor lighting 2.

RGB + IR with Y. The base YOLOv5 algorithm uses only RGB images, only IR images, or combines both RGB and IR images without using the SACAF module. Additionally, the experiment introduces an adaptive fusion module, RGB + IR + SACAF with Y, on top of the bimodal fusion. This comparison aims to highlight the advantages of multimodal fusion over unimodal fusion and the benefits of the proposed adaptive feature fusion module in multimodal target detection.

Table II demonstrates the recognition performance of both unimodal and multimodal networks. The logarithmic mean leakage rate (MR^{-2}) is employed as a metric for evaluating multimodal pedestrian detection performance. The AP is another key indicator, with a higher AP value indicating enhanced target detection performance.

Table II analysis shows that the bimodal fusion network (RGB + IR with Y) outperforms the unimodal (RGB with Y, IR with Y) in the comparison experiments. This result further confirms the effectiveness of bimodal feature fusion in enhancing target recognition capability. However, while bimodal fusion combines the features of both modalities, its fusion strategy is relatively simple, involving mainly point-by-point summation of the feature maps. This approach does not fully exploit the more significant and complementary features between the two modalities.

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TABLE II
COMPARISON OF RECOGNITION PERFORMANCE OF UNIMODAL AND MULTIMODAL NETWORKS

Mould	RGB	IR	SACAF	MR^{-2}/%	mAP0.5/%
RGB with Y	✓			26.5	78.5
IR with Y		✓		26.1	78.9
RGB+IR with Y	✓	✓		25.4	80.1
RGB+IR+SACAF with Y	✓	✓	✓	**21.9**	**83.6**

TABLE III
COMPARISON OF DIFFERENT ALGORITHMS FOR AP ON LLVIP DATASET

Differentiation	Datasets	mAP50/%	mAP%
Yolov5	RGB	90.8	78.5
Yolov5	IR	94.6	61.9
CFT [61]	RGB+IR	97.5	63.6
Infusion-Net [64]	RGB+IR	**98.6**	64.6
LRAF-Net [65]	RGB+IR	97.9	66.3
CAMF [66]	RGB+IR	89.01	55.6
Ours	RGB+IR	97.9	**68.1**

TABLE IV
COMPARISON OF LEAKAGE RATE OF DIFFERENT ALGORITHMS ON KAIST DATASET

Differentiation	All day MR^{-2}	Daytime MR^{-2}	Evening MR^{-2}
ACF [53]	47.32	42.57	56.15
Halfway Fusion [56]	25.75	24.88	26.59
SSD-Vis [67]	18.33	19.06	17.78
Fusion-RPN [62]	18.29	19.57	16.27
IAF-RCNN [68]	15.73	14.55	18.26
SSD-Multiscale [67]	15.66	17.85	11.39
MSDS-RCNN [57]	11.63	10.63	13.73
IPWF+ECA [67]	11.16	12.40	8.46
Ours	**9.79**	**9.55**	**9.93**

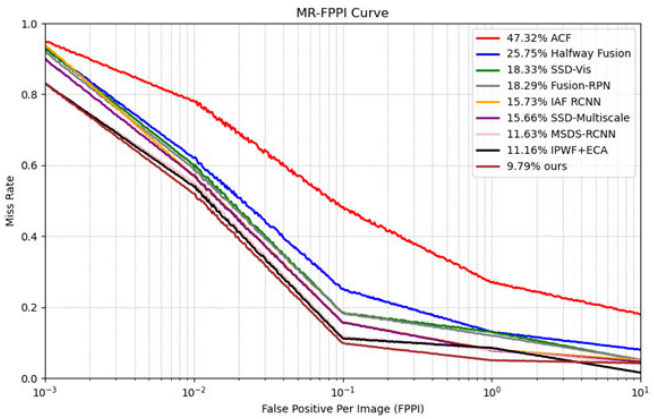


Fig. 9. MR-FPPI curves of different multimodal algorithms on the KAIST dataset.

To further improve recognition, the network introduces the SACAF module for experimental comparison. The experimental results indicate that the SACAF module significantly improves performance compared to the simple fusion of unimodal and traditional bimodal modalities.

Fig. 7 shows the $P-R$ curves for both the unimodal algorithm and the bimodal fusion algorithm. The input for the IR image is shown in Fig. 7(a) and the input for the RGB image is shown in Fig. 7(b). The algorithm used in this article is shown in Fig. 7(c). Experimental verification has shown that RGB and IR features have complementary advantages for different targets and scenes. The figure illustrates that the model which combines the two images has a higher AP value and improved accuracy for each category.

This fusion module enhances the model's recognition ability when dealing with complex and subtle targets by intelligently and flexibly integrating features from different modalities. Fig. 8 displays some of the test results.

The comparison between Fig. 8(a) and (b) reveals that well-lit daytime conditions result in extremely rare misdetections and omissions in RGB images. In contrast, IR images exhibit a certain degree of misdetection under such conditions due to the lack of detailed texture and rich color information. However, the application of a simple bimodal fusion network corrected some of the misdetections. However, the basic fusion method still results in omissions and false detections. This is mainly due to ineffective filtering of noise information, which is then added to the new feature maps, causing detection issues. After introducing the adaptive fusion module, the network model significantly reduces omissions and false detections. This is comparable to the detection results achieved using RGB images alone, proving the model's effectiveness even in well-lit environments.

However, as shown in Fig. 8(c) and (d), the situation is different in low-light conditions. Poor lighting conditions at night make it difficult to recognize targets in RGB images. In contrast, IR images provide clear contour information due to their advantages in low-light conditions, which is particularly important for nighttime target detection. In this case, the bimodal fusion network once again plays a crucial role, particularly with the inclusion of the adaptive fusion module. This module intelligently balances and integrates information from both modalities, significantly enhancing the network's ability to detect targets at night. This fusion strategy not only improves the accuracy of nighttime target detection but also reduces false detections and leakage, demonstrating excellent adaptability and effectiveness under varying lighting conditions.

In summary, the use of the adaptive fusion module demonstrates significant advantages in target detection under various lighting conditions. It can effectively improve detection performance during both daytime and nighttime, when there is sufficient or weak light, respectively. This is of great practical significance.

D. Experiments With Different Algorithms

To validate the robustness and broad applicability of this algorithm, we tested not only the aerial photography dataset but also the publicly available KAIST and LLVIP datasets. Although these two datasets mainly focus on pedestrian recognition, they also contain a small percentage of targets, providing an ideal test environment to verify the algorithm's ability to handle small targets.

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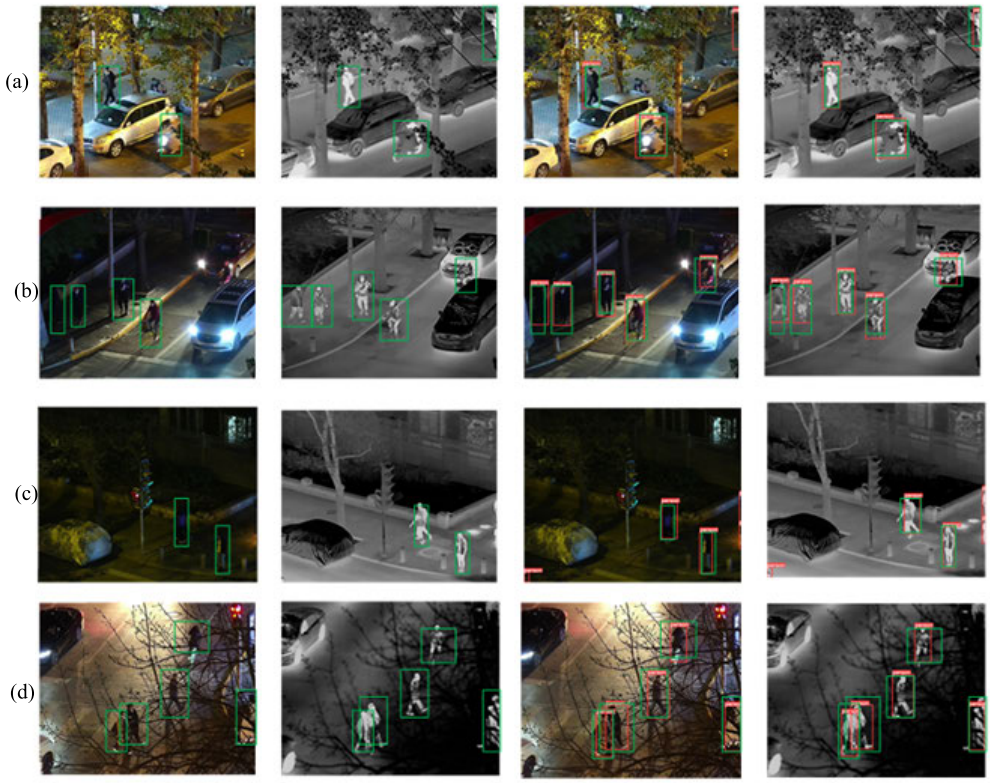


Fig. 10. LLVIP dataset test plots. The four groups of images (a)–(d) are, from left to right, the original RGB labeled graphs, the original IR labeled graphs, the output of the merged detection results in RGB, and the output of the merged detection results in IR.



Fig. 11. KAIST dataset test plots. The four groups of images (a)–(d) are, from left to right, the original RGB labeled graphs, the original IR labeled graphs, the output of the merged detection results in RGB, and the output of the merged detection results in IR.

Table III presents a detailed comparison of the performance of various algorithms on the two core performance metrics, mAP50 and mAP. The results indicate that the bimodal model-based approach exhibits a significant improvement in both metrics compared to the YOLOv5-based unimodal dataset. Our algorithm does not differ much from

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the LRAF-Net model on mAP50, while achieving a 1.8% improvement on mAP. Additionally, our algorithm outperforms the current best performing Infusion-Net detection network by 3.5% in mAP. The results of the study demonstrate that the algorithm incorporating the cross-attention module provides a more significant improvement in accuracy than the algorithm using light perception weights alone.

These experimental results demonstrate the superior performance of our algorithm on the LLVIP dataset and its better generalization ability.

Table IV and Fig. 9 present a comparison of the leakage rates of multiple multimodal algorithms on the KAIST dataset. The algorithms presented demonstrate excellent performance in both daytime and nighttime conditions and are capable of effectively recognizing pedestrian targets in images. Moreover, the leakage detection rate of the present algorithm is demonstrably lower than that of various other algorithms, irrespective of lighting conditions. This not only corroborates the efficacy and precision of the algorithms but also serves to reinforce their reliability and efficacy in practical applications.

E. Visualization of Results

Figs. 10 and 11 visually demonstrate the effectiveness of our algorithms tested on the LLVIP dataset and the KAIST dataset. Where the green box indicates the actual labeled box and the red box is the detection box. The detection category is indicated on the box by a marking. The scene in Fig. 10 shows the significant advantages of the multimodal fusion technique when dealing with complex environments for pedestrian detection tasks.

In Fig. 10(a), the upper-right figure is nearly invisible in the RGB image, but the IR image provides subtle silhouette information that, when effectively combined with the features of both images, allows for accurate target detection. In Fig. 10(b), the two pedestrians near the utility poles are difficult to distinguish from the background in the RGB image due to the dimly lit environment. However, by combining the IR image, their location can be determined more accurately. Fig. 10(c) demonstrates a scenario where the two figures on the right lack recognizable features in the RGB image but retain some silhouette information in the IR image, enabling accurate identification of their locations. In the scene depicted in Fig. 10(d), the tree branches cause significant occlusion of the figures, making the portraits almost unrecognizable in the RGB image. However, the IR image retains contour information that distinguishes the figures from the background noise, effectively assisting in target detection.

While the images above do contain some smaller, more microscopic targets, these can still be accurately identified. This finding confirms the advantage of our algorithm in achieving small target detection. This capability is critical when dealing with tiny or distant targets in complex scenes, demonstrating the algorithm's efficiency and accuracy in capturing subtle features and handling difficult recognition tasks.

The KAIST dataset contains a significant number of small targets with dimensions smaller than 30 $\times$ 30 pixels, which meets the definition of small targets. Fig. 11 shows that our algorithm accurately labels the location of the targets even under weak illumination and when only a small percentage of targets are present. While some targets were missed, the majority were recognized reliably. The algorithm's robustness is a significant advantage, ensuring system stability and reliability in practical applications. Further optimization and improvement are expected to reduce the leakage rate, enabling the algorithm to perform well in various complex environments.

V. Conclusion

This article introduces a YOLO-SA network module tailored for multimodal and adaptive feature fusion in airborne small target detection algorithms. The network consists of a modified YOLO network and the SACAF module. The SACAF module exhibits promising performance in detecting small targets. Feature extraction is enhanced through the utilization of two mechanisms: the spatial attention mechanism and the channel attention mechanism, operating synergistically. Additionally, an adaptive weighting formula is proposed to amplify valid features while suppressing irrelevant ones. The algorithm demonstrates effectiveness in identifying and precisely localizing small targets across diverse lighting conditions, particularly in low-light environments. Extensive evaluations on aerial photography datasets, including the KAIST and LLVIP datasets, illustrate the algorithm's robustness and significant advantages in detecting small target images. Although our algorithm performs well in terms of leakage rate and detection accuracy, the introduction of different attention modules leads to a reduction in the detection rate. In the future, methods to maintain high accuracy while reducing model parameters can be sought to further improve the detection rate of the network.

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